unmelt; memories are of the past, never of the future; people age, but they don't unage. These asymmetries govern our lives; the distinction between forward and backward in time is a prevailing element of experiential reality. If forward and backward in time exhibited the same symmetry we witness between left and right, or back and forth, the world would be unrecognizable. Eggs would unbreak as often as they broke; candles would unmelt as often as they melted; we'd remember as much about the future as we do about the past; people would unage as often as they aged. Certainly, such a time-symmetric reality is not our reality. But where does time's asymmetry come from? What is responsible for this most basic of all time's properties?

It turns out that the known and accepted laws of physics show no such asymmetry (Chapter 6): each direction in time, forward and backward, is treated by the laws without distinction. *And that's the origin of a huge puzzle.*Nothing in the equations of fundamental physics shows any sign of treating one direction in time differently from the other, and that is totally at odds with everything we experience.[5]

Surprisingly, even though we are focusing on a familiar feature of everyday life, the most convincing resolution of this mismatch between fundamental physics and basic experience requires us to contemplate the most unfamiliar of events—the beginning of the universe. This realization has its roots in the work of the great nineteenth-century physicist Ludwig Boltzmann, and in the years since has been elaborated on by many researchers, most notably the British mathematician Roger Penrose. As we will see, special physical conditions at the universe's inception (a highly ordered environment at or just after the big bang) may have imprinted a direction on time, rather as winding up a clock, twisting its spring into a highly ordered initial state, allows it to tick forward. Thus, in a sense we'll make precise, the breaking—as opposed to the unbreaking— of an egg bears witness to conditions at the birth of the universe some 14 billion years ago.

This unexpected link between everyday experience and the early universe provides insight into why events unfold one way in time and never the reverse, but it does not fully solve the mystery of time's arrow. Instead, it shifts the puzzle to the realm of *cosmology*—the study of the origin and evolution of the entire cosmos—and compels us to find out